

# A Generative AI Framework for Contractor Cost Analysis

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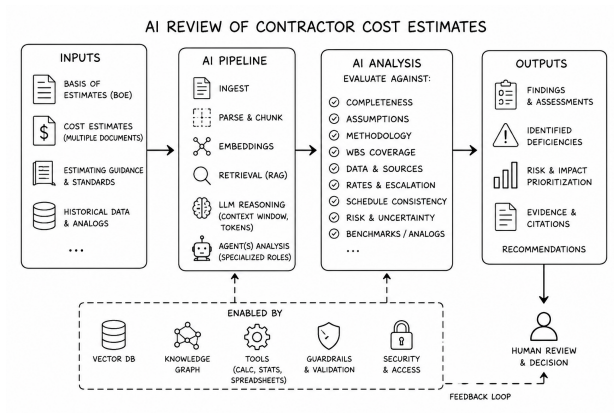
BAE Systems, Inc.

July 29, 2026

# Mission Statement

**What information** must be provided to a **generative AI** system to enable **credible evaluation** of **cost estimates**?

# Schematic



# Presentation Overview I

- 1 Preliminaries
  - Tasking
  - Personae Dramatis
  - Repository
  - Survey of AI Landscape
    - AI for Federal Cost Estimation
    - YouTube
  - Sampler of Cost Estimation Guidance
    - FHA
    - CBO
    - GAO
    - Commerce
    - Contractors

- 2 Inputs

# Presentation Overview II

3 AI Pipeline

4 AI Components

5 Outputs

# Where Does This Research Stand?

1. Tasking
2. Online Library
3. Survey of AI Landscape

# Collection of Tasking Documents

emails, memos, phone calls, ...

# Primary Players

## 1. invasion



# GitLab Repo

## Best Practice: Organize data in GitLab

# GitLab Repo Example I

The screenshot shows the GitLab repository page for 'spack/spack'. The repository is public and has 212 branches and 215 tags. The 'Code' tab is selected, showing a list of files and their commit history. The files listed are .ci, .devcontainer, .github, bin, etc/spack, lib/spack, share/spack, var/spack, and .codecov.yml. Each file entry includes a commit message and a timestamp.

| File          | Commit Message                                              | Timestamp    |
|---------------|-------------------------------------------------------------|--------------|
| .ci           | CI: Point at the latest hidden snapshot ref (#52540)        | last month   |
| .devcontainer | codespaces: add ubuntu22.04 (#46100)                        | 2 years ago  |
| .github       | gha: swap os<->python-version in unit-tests (#52763)        | last week    |
| bin           | Windows Shells: Allow bin/spack to work as expected (#52... | 2 months ago |
| etc/spack     | Allow multiple py-setuptools-scm specs in a dependency ...  | last month   |
| lib/spack     | spec.py: remove finalize branch in _constrain (#52778)      | yesterday    |
| share/spack   | Bugfix: Add missing config view to Mirror update (#52491)   | yesterday    |
| var/spack     | builtin_mock: use C sources in garply/quux/corge (#52719)   | 2 weeks ago  |
| .codecov.yml  | refactor: move vendored dependencies under spack modu...    | last year    |

# GitLab Repo Example II

README Code of conduct Contributing Apache-2.0 license MIT license Security



# Spack

ci passing

Bootstrapping passing

Containers passing

docs passing

codecov 84%

slack 4207

matrix 50 users

[Getting Started](#) • [Config](#) • [Community](#) • [Contributing](#) • [Packaging Guide](#) • [Packages](#)

Spack is a multi-platform package manager that builds and installs multiple versions and configurations of software. It works on Linux, macOS, Windows, and many supercomputers. Spack is non-destructive: installing a new version of a package does not break existing installations, so many configurations of the same package can coexist.

Spack offers a simple "spec" syntax that allows users to specify versions and configuration options. Package files are written in pure Python, and specs allow package authors to write a single script for many different builds of the same package. With Spack, you can build your software *all* the ways you want to.

See the [Feature Overview](#) for examples and highlights.

## Plan B: Schema

```
p:\waynes-world/  
├ components  
├ inputs  
├ outputs  
├ pipeline  
└ tasking
```

# Directory vs. Repository

| Directory          | Repository               |
|--------------------|--------------------------|
| Siloed, local only | Shared, collaborative    |
| No history         | Full audit trail         |
| No branching       | Unlimited branches       |
| Manual backups     | Automated remotes        |
| No access control  | Fine-grained permissions |
| No tools           | Full dev toolchain       |
| Risk of loss       | Backup by design         |

***A plain directory is just storage. A repository is a living system for building software and cataloging results.***

# Where Does This Research Belong?



# Where Does This Research Belong?

- Can machine learning produce better estimates?
- Can AI automate portions of the estimating workflow?
- Can generative AI help an estimator prepare a basis of estimate?
- Can AI retrieve analogous historical programs or identify cost drivers?

*What **AI architecture** and **domain knowledge** are required to evaluate a collection of cost estimates credibly?*

# AI for Federal Cost Estimation and Analysis

## AI for Federal Cost Estimation and Analysis Course

ID TECH9036

Hands-on Training from Experts

Live Online or Self-Paced

Apply AI tools to strengthen cost estimation, economic analysis, and capital planning workflows while maintaining compliance with GAO and OMB standards.

View Schedule (Live Online) ⌚

Book Now (Self-Paced) >

### This course includes

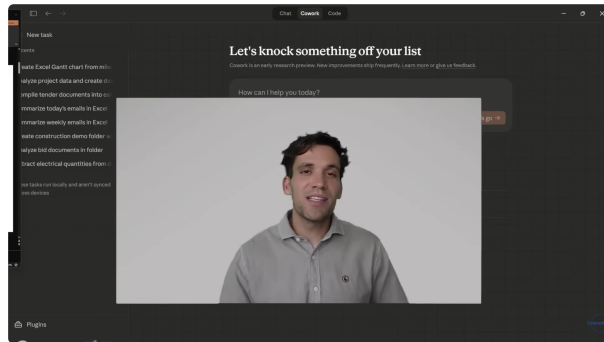
- ⌚ 4 hours of live, project-based training from experts
- 📄 Verified digital certificate of completion
- 📄 Proprietary workbook included
- 📄 Accredited program
- 🔄 Retake for free within 1 year
- 📄 Credits: 0.4 CEUs



# AI for Federal Cost Estimation and Analysis

*This half-day course teaches federal cost professionals how to use **AI as a practical tool** for developing, **reviewing**, and communicating **cost estimates** and economic analyses. Participants will learn to **apply AI** to lifecycle cost modeling, cost-benefit and cost-effectiveness analysis, **independent cost estimate review**, and capital planning — while maintaining compliance with GAO Cost Estimating and Assessment Guide standards and OMB requirements.*

# Claude Cowork for Construction - Estimating



## Claude Cowork for Construction - Estimating, Contract Management and Scheduling



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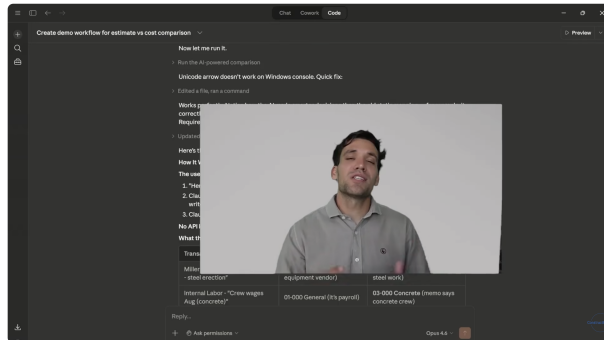
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Claude Code for Construction - Estimating, Cost Control and Construction Workflows



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# Major Project Program Cost Estimating Guidance

 U.S. Department of Transportation  
Federal Highway Administration

AboutProgramsResourcesBriefing

## Major Projects

Cost EstimatingFinancial PlansProject Management PlansProject Delivery Lessons Learned

ProcessGuidance

Home / Programs / Major Projects / Cost Estimating / Guidance

### Guidance: Major Project Program Cost Estimating Guidance

(January 2007)

Estimates are central to establishing the basis for key project decisions, for establishing the metrics against which project success will be measured and for communicating the status of a project at any given point in time. Logical and reasonable cost estimates are necessary in maintaining public confidence and trust throughout the life of a major project.

This guidance is for the preparation of a total program cost estimate for a major project. For the purpose of this guidance, a major project is defined as a project that receives any amount of Federal financial assistance and has an estimated total program cost greater than \$500 million (expressed in year-of-expenditure dollars), or other projects identified as a major project by the FHWA. The total program cost estimate includes construction, engineering, acquisition of right-of-way, and related costs, which will be identified by this guidance. Although this guidance is for major projects, it may also be applied to other projects.

Major projects by nature are usually more complex and contain more risk elements than other projects. Careful attention must be provided when preparing cost estimates for major projects. Traditional estimating methods may not be appropriate in all cases. This guidance is intended to assist State transportation departments, the FHWA, and other sponsoring agencies with ensuring that all program cost estimates are prepared using sound practices that result in logical and realistic initial estimated costs of the projects, providing a more stable cost estimate throughout the project continuum.

# Major Project Program Cost Estimating Guidance

Major Project Program Cost Estimating Guidance

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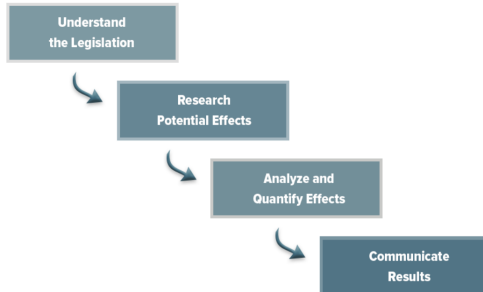
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# Congressional Budget Office

April 2023

## CBO Describes Its Cost-Estimating Process

### CBO's Four-Step Process for Preparing Cost Estimates

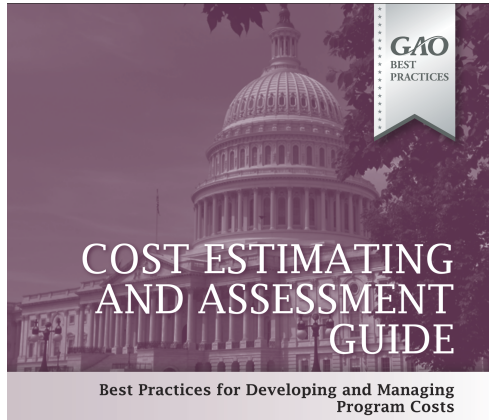


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# Government Accountability Office

## GAO COST ESTIMATING AND ASSESSMENT GUIDE

# Government Accountability Office: Best Practices





# Department of Commerce

## Department of Commerce Cost Estimating Guide

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Version 1.1

# Federal Pricing Group

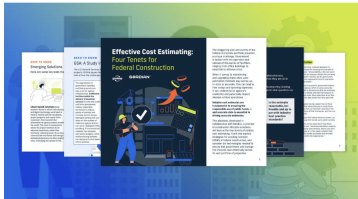


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## Cost/Price Analysis for Federal Contracting Officers

Federal Pricing Group provides cost and price analysis and cost estimating expertise to Federal Contracting Officers which enhances the Government's negotiating position and helps our Federal clients secure a better deal for the American taxpayer.

# Gordian



## Effective Federal Cost Estimation: Four Tenets of Reliable Estimates

October 8, 2024

 By Gordian

The staggering size and variety of the [federal](#) real estate portfolio presents a unique challenge. Government is tasked with the operation and upkeep of thousands of facilities, ranging from office buildings to hospitals to defense sites.

# Input to AI

1. Contractor estimates
2. Historical estimates
3. Cost-estimating handbook
4. Program documentation
5. Acquisition regulations

# AI Pipeline

1. Document ingestion
2. Parsing
3. Chunking
4. Embedding
5. Retrieval
6. Prompt construction
7. LLM reasoning
8. Report generation

# AI Components

1. Retrieval pipeline (RAG)
2. Agent(s)
3. Prompt templates
4. Token budgeting
5. Context window management
6. Human review

# Outputs

1. Findings
2. Missing assumptions
3. Inconsistencies
4. Confidence
5. Evidence citations

# References I



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